

RAJALAKSHMI INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

**AI-Assisted Condition Monitoring of Industrial
Gearbox Systems Using Unsupervised Anomaly
Detection**

Bachelor's Thesis

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Abstract

Industrial gearbox systems are critical components in rotating machinery environments and require continuous monitoring to reduce unexpected failures and operational downtime. Traditional condition monitoring systems commonly rely on threshold-based monitoring and expert interpretation of telemetry signals. However, the increasing availability of industrial telemetry data enables the integration of data-driven anomaly detection approaches into industrial monitoring workflows.

This thesis investigates the use of unsupervised anomaly detection methods for AI-assisted condition monitoring of industrial gearbox systems. The work focuses on telemetry analysis using vibration RMS measurements, frequency-domain spectrum analysis, operational telemetry signals, and dashboard-assisted monitoring workflows. Since labelled industrial failure datasets are often limited, the study emphasises unsupervised learning approaches capable of identifying deviations from normal operational behaviour without requiring extensive fault annotations.

A practical prototype workflow was developed using telemetry data collected in collaboration with Flender Drives Private Limited. The workflow included telemetry preprocessing, statistical feature extraction, FFT-based spectrum analysis, and Isolation Forest anomaly detection. The implementation was designed as an engineering-oriented proof-of-concept rather than a production-certified industrial diagnostic system.

The results indicate that telemetry-driven anomaly detection workflows can support industrial condition monitoring by assisting engineers in identifying unusual operational behaviour patterns through visualisation-assisted inspection and anomaly-oriented monitoring. However, additional long-term validation and deployment-oriented evaluation would be required for production-scale industrial adoption.

Keywords: *Condition Monitoring, Gearbox, Anomaly Detection, Isolation Forest, Predictive Maintenance, Vibration Analysis, Telemetry, Unsupervised Learning*

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1. Introduction

1.1 Motivation

Industrial gearboxes are among the most mechanically demanding components in modern manufacturing, energy, and process automation environments. These systems transfer power between rotating shafts, often operating under high torques, variable loads, and continuous duty cycles over years of service. A failure in a gearbox can cascade into unplanned production shutdowns, costly emergency maintenance, secondary equipment damage, and safety incidents. In large-scale industrial facilities, the economic consequences of unplanned gearbox downtime can reach hundreds of thousands of dollars per hour [1].

Historically, gearbox maintenance has been conducted on fixed-time schedules, replacing components according to manufacturer recommendations regardless of actual equipment condition. While this strategy prevents some catastrophic failures, it results in the premature replacement of healthy components and does not account for the heterogeneous operating conditions present in real industrial deployments. Condition-based maintenance emerged as an improvement, tying maintenance actions to observable indicators of equipment degradation such as vibration amplitude, temperature, and lubrication condition [2].

The rapid proliferation of industrial IoT (IIoT) sensors and condition monitoring platforms has created new opportunities to collect high-frequency telemetry data directly from operational gearboxes. Platforms such as the Flender CONDITION ANALYZER provide continuous access to vibration, temperature, and speed signals sampled at thousands of data points per second. However, the sheer volume and dimensionality of this telemetry data creates a significant challenge: meaningful interpretation of signals requires either expert domain knowledge or automated analytics capable of separating normal operational variation from early-stage fault indicators [3].

Machine learning offers a compelling pathway to automate the detection of anomalous behaviour in industrial telemetry streams. Unsupervised learning methods are particularly well-suited to this domain because industrial machinery rarely accumulates labelled failure event datasets of sufficient size to train supervised

classifiers. Isolation Forest, a tree-ensemble anomaly detection algorithm, has demonstrated favourable performance in high-dimensional, partially contaminated datasets and is well-suited to the characteristics of industrial gearbox telemetry [4].

This thesis was developed in collaboration with Flender Drives Private Limited, a leading manufacturer of industrial drive systems, providing access to real operational telemetry from a monitored gearbox installation. This industry collaboration enabled the development and evaluation of a realistic proof-of-concept anomaly detection workflow grounded in actual operational data.

1.2 Problem Statement

The central problem addressed in this thesis is the detection of abnormal operational behaviour in industrial gearbox systems using telemetry data, under conditions where labelled fault examples are unavailable or insufficient for supervised learning. Current industrial monitoring platforms generate rich telemetry streams — including vibration RMS, oil temperature, and input speed — that contain embedded information about equipment health. However, extracting actionable anomaly indicators from these streams without expert manual analysis represents a significant open challenge.

Threshold-based alarm systems, while simple to implement, suffer from two fundamental limitations. First, they require a priori knowledge of fault severity thresholds, which are often conservatively set to avoid false alarms at the cost of delayed detection. Second, they treat each telemetry channel independently, failing to capture multi-variate interactions between signals that may indicate emerging faults before any single parameter crosses a predefined limit [5].

Data-driven approaches can potentially overcome these limitations by learning normal operational patterns from historical data and flagging multivariate deviations as anomalous. However, applying machine learning to industrial telemetry introduces practical challenges including signal non-stationarity, operational regime changes (load, speed), sensor noise, and the absence of labelled training examples for rare fault events.

1.3 Research Question

The central research question guiding this thesis is:

"Can unsupervised anomaly detection methods support identification of abnormal operational behaviour in industrial gearbox telemetry environments where labelled fault data is limited?"

Supporting sub-questions include: (i) Which telemetry features extracted from vibration, temperature, and speed signals are most informative for anomaly detection? (ii) How does Isolation Forest perform on multi-channel industrial gearbox telemetry? (iii) What are the practical limitations of unsupervised anomaly detection in real industrial deployments?

1.4 Objectives

The specific objectives of this thesis are as follows:

- To survey existing literature on condition monitoring, predictive maintenance, and machine learning approaches for industrial equipment fault detection.
- To characterise the telemetry data available from the Flender CONDITION ANALYZER monitoring platform, including vibration RMS, oil temperature, and input speed signals.
- To develop a telemetry preprocessing and feature engineering pipeline suited to high-frequency industrial vibration data.
- To implement and configure an Isolation Forest anomaly detection model for unsupervised anomaly identification in multi-channel gearbox telemetry.
- To evaluate the anomaly detection prototype through visualisation-assisted analysis and engineering interpretation of flagged events.
- To identify practical limitations and propose directions for future research and system improvement.

1.5 Scope and Limitations

This thesis is scoped as an engineering-oriented proof-of-concept study. The prototype system was developed and evaluated using telemetry data from a single monitored gearbox installation in collaboration with Flender Drives Private Limited. The following scope boundaries apply:

- The study focuses on unsupervised anomaly detection; fault classification and root cause diagnosis are outside scope.

- Only the X-axis vibration signals, oil temperature, and input speed telemetry channels are analysed; multi-axis fusion and advanced bearing-specific KPIs (inner ring, roller, outer ring) are not fully explored.
- The prototype is not designed or validated for deployment in safety-critical or production-certified industrial systems.
- Long-term validation over extended operational periods was not possible within the project timeline.
- Ground-truth fault labels were not available; evaluation relies on engineering inspection and domain expert interpretation.

2. Fundamentals and Background

2.1 Industrial Gearbox Systems

An industrial gearbox is a mechanical power transmission device that modifies rotational speed and torque between a prime mover (typically an electric motor) and a driven load. Gearboxes are classified by their gear arrangement into helical, bevel, planetary, worm, and multi-stage configurations depending on the required speed ratio, power rating, and physical footprint of the application [6].

Key mechanical components of a gearbox include the input and output shafts, bearing assemblies supporting each shaft, gear wheels forming meshing pairs, lubrication systems (oil bath or forced circulation), and the housing enclosure. Each of these components generates characteristic signatures in the vibration spectrum during normal operation, and deviates from these signatures in response to developing faults [7].

Common gearbox fault modes include: gear tooth wear and micropitting, bearing race fatigue (pitting on inner race, outer race, or rolling elements), lubrication degradation, shaft misalignment, rotor imbalance, and housing resonance excitation. Each fault mode produces identifiable spectral features at frequencies related to shaft rotation rate, gear mesh frequency, and bearing defect frequencies. For example, gear mesh frequency (GMF) is defined as the product of shaft rotational frequency and the number of gear teeth, while bearing defect frequencies are computed from bearing geometry parameters and shaft speed [8].

2.2 Condition Monitoring

Condition monitoring (CM) refers to the continuous or periodic measurement and assessment of parameters that indicate the health of operating machinery. The objective of condition monitoring is to detect deterioration at an early stage, enabling maintenance planning before functional failure occurs [9].

The primary techniques employed in gearbox condition monitoring are:

- **Vibration Analysis:** Accelerometers mounted on gearbox housings capture the dynamic response of the structure to internal forcing mechanisms. Both time-domain statistics (RMS, peak, kurtosis, crest factor) and frequency-domain

features (spectral amplitude at characteristic frequencies) are used as condition indicators.

- **Oil and Debris Analysis:** Particle counting and spectroscopic analysis of lubricant samples can identify wear debris generated by gear and bearing surfaces. This technique provides insight into wear mechanisms but requires periodic manual sampling.
- **Thermography and Temperature Monitoring:** Elevated oil and bearing temperatures can indicate inadequate lubrication, excessive friction, or overloading. Oil temperature sensors are standard instrumentation on monitored gearboxes.
- **Acoustic Emission:** High-frequency acoustic signals (100 kHz–1 MHz) generated by crack propagation and surface fatigue can provide early-stage fault indicators, though signal interpretation requires specialised expertise.

Modern industrial CM platforms integrate sensor acquisition, signal processing, and alarm management into a unified software environment. The Flender CONDITION ANALYZER used in this study exemplifies this architecture, providing broadband KPI calculation, shaft-specific diagnostic indices, and bearing defect frequency monitoring through a dashboard interface [10].

2.3 Predictive Maintenance

Predictive maintenance (PdM) represents a condition monitoring strategy in which maintenance actions are triggered by indicators of impending failure rather than elapsed time or fixed schedules. PdM seeks to maximise equipment availability by intervening at the optimal point in a degradation trajectory — late enough to avoid premature replacement of healthy components, but early enough to prevent unexpected failure [11].

The implementation of PdM requires three core capabilities: (i) continuous or high-frequency monitoring of relevant condition indicators; (ii) a prognostic model that can estimate remaining useful life (RUL) or flag anomalous behaviour; and (iii) an integration with maintenance planning systems to translate condition assessments into maintenance work orders. Data-driven PdM methods have gained significant research attention due to their ability to learn degradation patterns from historical operational data without requiring detailed physics-based models of complex failure mechanisms [12].

2.4 Telemetry Data

In the context of this thesis, telemetry data refers to the digitised measurements continuously streamed from sensors installed on the monitored gearbox. The Flender CONDITION ANALYZER acquisition system sampled vibration signals at 13,333 Hz (samples per second), providing high temporal resolution suitable for frequency-domain analysis up to approximately 6,667 Hz (Nyquist limit). The primary telemetry channels used in this study are:

Signal Channel	Sensor Type	Physical Unit	Sampling Rate
Vibration RMS (X-axis)	Accelerometer	m/s ²	13,333 Hz
Input Speed	Speed sensor / Accelerometer	m/s ² / RPM	13,333 Hz
Oil Temperature	Thermocouple / RTD	°C	Low-rate
Broadband KPIs	Derived from accelerometer	Dimensionless	Computed

Table 1 – Summary of telemetry channels available from the Flender CONDITION ANALYZER monitoring platform.

The raw vibration measurements captured from the monitored gearbox system at X-axis were used as the primary dataset for prototype development and analysis.

2.5 Vibration Analysis

Vibration analysis is the most widely applied technique for gearbox condition monitoring due to the information-rich nature of vibration signals. Vibration in a gearbox is generated by the periodic meshing of gear tooth pairs, by bearing rolling contact, and by any structural or balance asymmetries in rotating components [13].

Time-domain analysis extracts statistical descriptors from the raw acceleration waveform. The Root Mean Square (RMS) value measures the effective energy content of the signal and is a standard broadband health indicator: an increase in vibration RMS above baseline levels suggests increasing dynamic forces within the gearbox. Peak value and crest factor (peak-to-RMS ratio) are sensitive to impulsive events such as bearing defect impacts. Kurtosis quantifies the degree of impulsiveness in the signal distribution and is particularly sensitive to early-stage bearing fatigue where discrete impact events are superimposed on the background vibration [14].

A key observation from the raw vibration data collected in this study (Figure 2 : Vibration RMS Raw Data) is that the signal exhibits a relatively stationary amplitude envelope with rapid oscillations in the $\pm 0.4 \text{ m/s}^2$ throughout the 2-second acquisition window. This behaviour is characteristic of steady-state operation at constant load and speed, providing a suitable baseline for anomaly detection model training.

2.6 Frequency Spectrum Analysis

Frequency-domain analysis transforms the time-domain vibration waveform into its constituent frequency components using the Fast Fourier Transform (FFT). The resulting power spectrum identifies the frequencies at which significant vibration energy is concentrated, enabling identification of rotating machinery characteristic frequencies [15].

For a gearbox operating at rotational frequency f_r (Hz) with Z gear teeth, the gear mesh frequency is $f_{\text{GMF}} = f_r \times Z$. Harmonics of f_{GMF} ($2 \times f_{\text{GMF}}$, $3 \times f_{\text{GMF}}$) and sidebands spaced at shaft frequency intervals around f_{GMF} are associated with specific fault modes including tooth wear, eccentricity, and load variation. Bearing defect frequencies (ball pass frequency outer race BPFO, ball pass frequency inner race BPFI, ball spin frequency BSF) are calculated from bearing geometry and rotational speed, and elevated spectral energy at these frequencies is indicative of race or rolling element fatigue [16].

The frequency spectrum visualisation from the monitoring dashboard (Figure 1: Vibration RMS Spectrum) shows the vibration energy distribution from 0 Hz to approximately 6,700 Hz. A dominant peak is visible in the low-frequency region (below 200 Hz), consistent with fundamental shaft rotation and low-order gear mesh harmonics. A secondary spectral feature appears around 3,500–3,700 Hz, which merits investigation against bearing defect frequency calculations. The spectrum is otherwise characterised by a low noise floor, indicating clean signal acquisition and absence of broadband elevated noise that would suggest advanced degradation.



Figure 1 – Frequency-domain spectrum visualisation from the industrial gearbox monitoring dashboard (Vibration RMS, Spectrum view).

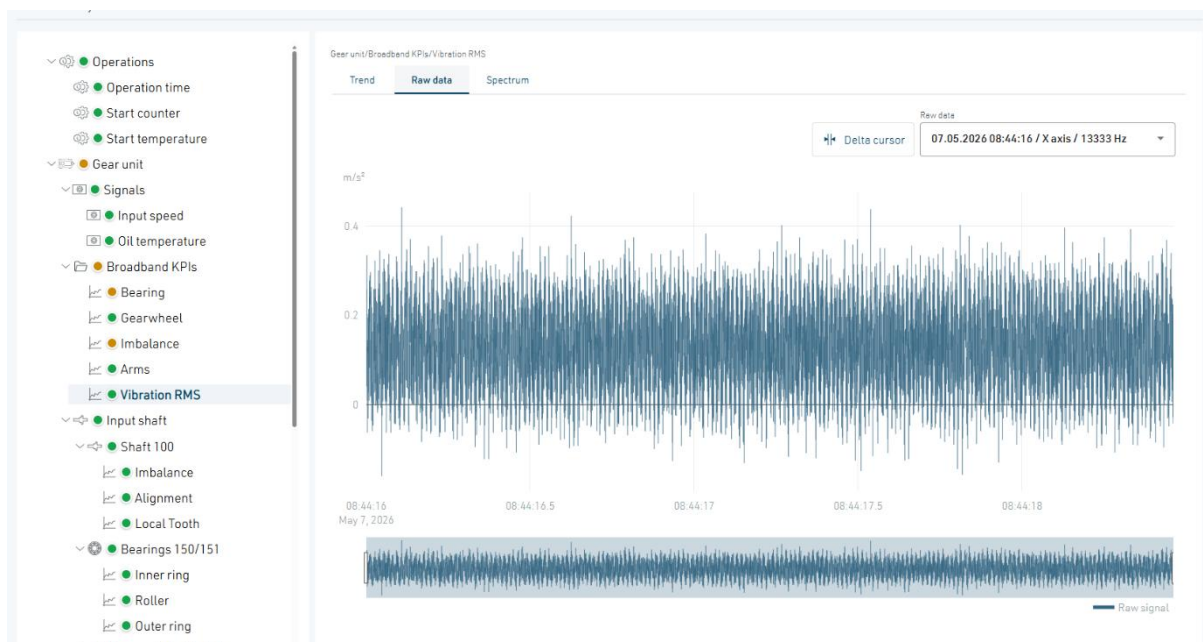


Figure 2 – Raw vibration time-domain signal from Vibration RMS monitoring (2-second acquisition window).

3. Literature Review

3.1 Traditional Monitoring Approaches

The earliest gearbox monitoring systems relied on periodic manual inspection, oil sampling, and time-domain vibration measurements using portable data collectors. Scheffer and Girdhar [17] provide a comprehensive treatment of traditional vibration monitoring practice, including the use of ISO vibration severity charts (ISO 10816) to classify machine condition on the basis of broadband vibration RMS values. While these standards provide useful overall condition classifications, they do not differentiate between fault-specific indicators and operationally induced vibration changes.

Threshold-based alarm systems represent the dominant implementation approach in deployed industrial monitoring platforms. A fixed upper threshold is defined for each monitored parameter (vibration RMS, temperature, KPI), and an alarm is triggered when the measurement exceeds this limit. Lei et al. [18] note that fixed-threshold systems are prone to both false alarms (triggered by transient operational events such as load changes) and missed detections (when gradual degradation remains below the alarm threshold until catastrophic failure is imminent). Statistical control chart approaches (e.g., Shewhart X-bar charts, CUSUM charts) have been proposed as improvements that account for natural process variability, but these require the definition of in-control reference distributions and are still univariate in nature.

Gear mesh frequency tracking and order tracking methods represent a step forward in specificity, isolating vibration energy associated with particular mechanical events rather than evaluating broadband signal energy. Randall [19] provides a detailed theoretical treatment of gear and bearing diagnostic techniques based on spectral analysis, including cepstrum analysis for separating excitation sources. However, these methods require precise knowledge of shaft rotational speed, gear tooth counts, and bearing geometry parameters, which may not always be available or accurately specified in legacy installations.

3.2 Machine Learning for Predictive Maintenance

The application of machine learning to machinery fault detection and prognostics has been an active research area since the late 1990s, gaining significant momentum with

the availability of labelled bearing and gearbox datasets such as the CWRU Bearing Dataset, the Paderborn University Bearing Dataset, and the PRONOSTIA bearing run-to-failure dataset [20].

Early ML applications in this domain relied primarily on supervised classification models. Widodo and Yang [21] conducted a comprehensive survey of Support Vector Machine (SVM) applications to rotating machinery fault diagnosis, demonstrating that SVM classifiers trained on time-frequency features extracted from vibration signals can achieve high classification accuracy for distinguishing healthy and faulty bearing conditions under fixed operating conditions.

Deep learning approaches have extended the representational capacity of fault detection models. Zhao et al. [22] demonstrated that convolutional neural networks (CNNs) applied directly to raw vibration waveforms or spectrogram images can learn hierarchical fault-discriminative features without manual feature engineering. Recurrent architectures (LSTM networks) have been applied to remaining useful life prediction by modelling the temporal evolution of degradation indicators [23].

More recent work has explored transformer-based attention mechanisms for vibration signal analysis. Li et al. [24] proposed a multi-head self-attention architecture for bearing fault diagnosis, demonstrating superior cross-domain generalisation compared to CNN baselines. However, these advanced supervised methods all share the fundamental requirement of labelled training data encompassing representative examples of the fault modes to be detected.

3.3 Supervised vs. Unsupervised Learning

The dichotomy between supervised and unsupervised learning is particularly consequential in industrial fault detection. Supervised methods offer the highest diagnostic accuracy when sufficient labelled training data is available and the test conditions match the training distribution. In practice, however, labelled fault datasets for specific industrial assets are rarely available at the scale required for robust supervised model training. The expense of deliberately inducing controlled faults for data collection, combined with the long incubation periods of fatigue-based failure mechanisms, makes supervised approaches difficult to implement in many real industrial deployments [25].

Unsupervised anomaly detection circumvents the labelling requirement by training models exclusively on data representing normal operational behaviour, then identifying test samples that deviate significantly from the learned normal distribution. This approach is well-aligned with the industrial scenario: while historical failure data is scarce, abundant normal operational telemetry is routinely collected by condition monitoring systems. The key challenge in unsupervised approaches is controlling the false positive rate, since operational regime changes (speed transients, load variations) can generate legitimate non-anomalous deviations from the training distribution that may be incorrectly flagged as faults [26].

Semi-supervised methods represent a middle ground, training on normal data with the optional incorporation of a small number of labelled anomaly examples. One-Class SVM, autoencoders trained on normal data, and SVDD (Support Vector Data Description) are established semi-supervised approaches that have been applied to machinery fault detection with promising results [27].

3.4 Isolation Forest

Isolation Forest (iForest), proposed by Liu, Ting, and Zhou [4], is an ensemble anomaly detection algorithm based on the principle that anomalous observations are easier to isolate than normal observations in feature space. The algorithm constructs an ensemble of Isolation Trees (iTrees), each of which recursively partitions the feature space by randomly selecting a feature and a split value. Anomalies, being isolated in sparse regions of feature space, require fewer partitioning steps (shorter path lengths) to isolate, while normal observations require more steps because they occupy dense regions surrounded by other similar observations.

The anomaly score for an observation is computed as a normalised function of its average path length across the forest. Observations with anomaly scores approaching 1.0 are flagged as highly anomalous; observations with scores near 0.5 are indistinguishable from the expected distribution of path lengths for normal observations. Liu et al. demonstrated that iForest achieves competitive anomaly detection performance against one-class SVM and LOF (Local Outlier Factor) on several benchmark datasets while offering significantly lower computational complexity — linear in the number of training samples — making it well-suited to large-scale industrial telemetry datasets [4].

Several properties of Isolation Forest make it particularly attractive for industrial gearbox telemetry analysis: (i) it handles high-dimensional feature vectors effectively without the curse of dimensionality that affects distance-based methods; (ii) it is robust to partial feature irrelevance, as random feature selection implicitly downweights non-discriminative features; (iii) the contamination hyperparameter allows explicit control over the expected proportion of anomalous observations in the training set, accommodating the practical situation where training data may include some abnormal operating events.

Chen et al. [28] applied Isolation Forest to wind turbine SCADA data for anomaly detection, demonstrating the algorithm's effectiveness in detecting early-stage drivetrain anomalies prior to maintenance events. Zhao et al. [29] compared iForest with autoencoder-based anomaly detection on industrial process data, finding that iForest provided more consistent results on datasets with limited anomaly examples, consistent with the scenario in this thesis.

3.5 Research Gap

While the individual components of the proposed workflow — vibration analysis, FFT feature extraction, and Isolation Forest anomaly detection — are established in the research literature, their integration into a practical industrial monitoring prototype validated on real-world gearbox telemetry from a commercial condition monitoring platform remains underexplored. Most published studies use benchmark laboratory datasets (CWRU, Paderborn) collected under controlled conditions, limiting the generalisability of findings to operational industrial environments [30].

Furthermore, most ML-based condition monitoring studies focus on classification accuracy under laboratory fault scenarios rather than the more practically relevant question of whether unsupervised anomaly detection can provide useful operational decision support in the absence of labelled fault examples. This thesis addresses this gap by developing and evaluating a prototype system using real operational telemetry from a monitored industrial gearbox, with evaluation grounded in engineering interpretation rather than classification performance metrics.

4. Methodology

4.1 Data Collection Workflow

Telemetry data was collected from an industrial gearbox installation monitored by the Flender CONDITION ANALYZER platform. The CONDITION ANALYZER system acquires high-frequency vibration data from accelerometers mounted on the gearbox housing, along with oil temperature and input speed measurements. Data is stored internally and accessible through the platform's web-based monitoring dashboard, which provides trend, raw data, and spectrum views for each monitored signal channel.

For this study, a single measurement session captured from the monitored gearbox system was selected as the primary dataset. This measurement captured 2.5 seconds of raw vibration data at 13,333 Hz, yielding approximately 33,332 samples per channel. The monitoring system structure available during data collection included the following hierarchy:

- Operations: Operation time, Start counter, Start temperature
- Gear unit > Signals: Input speed, Oil temperature
- Gear unit > Broadband KPIs: Bearing, Gearwheel, Imbalance, Arms, Vibration RMS
- Input shaft > Shaft 100: Imbalance, Alignment, Local Tooth
- Input shaft > Bearings 150/151: Inner ring, Roller, Outer ring

The status indicator system (green/yellow circles in the dashboard navigation panel) provides a quick visual summary of condition assessment across monitored elements. At the time of data collection, most elements showed green (normal) status, with amber (warning) indicators noted for Bearing KPI, Imbalance KPI, and the Gear unit operational status, suggesting the system was in a low-alert but non-critical operational state.

4.2 Telemetry Signals

Three primary telemetry channels were selected for anomaly detection analysis based on their direct relevance to gearbox health assessment:

4.2.1 Vibration RMS

The Vibration RMS channel provides the broadband vibration energy of the gearbox as measured by the X-axis accelerometer. This signal represents the integrated dynamic behaviour of all internal mechanical excitation sources (gear mesh, bearing contacts, shaft rotation). It is the most widely used single-channel health indicator for gearbox monitoring and serves as the primary analysis channel in this study [31].

4.2.2 Oil Temperature

Oil temperature is a secondary condition indicator that reflects the thermal state of the lubrication system. Elevated oil temperature can indicate inadequate oil flow, deteriorated lubricant viscosity, or increased friction due to surface wear. Although oil temperature typically changes on a longer timescale than vibration signals (minutes to hours rather than milliseconds), trends in oil temperature measured over extended operational periods provide complementary health information [32].

4.2.3 Input Speed

The input speed signal captures the rotational speed of the input shaft, which is critical for normalising vibration measurements and computing order-domain features. Speed variations can directly influence vibration amplitude and frequency content, making speed measurement essential for distinguishing operationally induced signal changes from true fault-induced anomalies. The measured raw data showed peak-to-peak variations of approximately 0 to 0.4 m/s² amplitude at 13,333 Hz sampling rate.

4.3 Data Preprocessing

The following preprocessing steps were applied to the raw telemetry data prior to feature extraction:

Signal Quality Verification: Visual inspection of raw waveforms (as shown in the monitoring dashboard raw data views) confirmed the absence of clipping, dropout, and obvious sensor artefacts. The input speed and vibration RMS raw signals both exhibited stationary amplitude envelopes without abrupt discontinuities, confirming data integrity for the selected measurement window.

DC Offset Removal: Any constant offset in the acceleration signal was removed by subtracting the signal mean. This step is standard practice in vibration analysis to ensure that FFT computation is not dominated by a DC component [33].

Windowing: The 33,332-sample acquisition was divided into non-overlapping analysis windows of 4,096 samples (approximately 307 ms duration at 13,333 Hz sampling rate). This window length provides a frequency resolution of approximately 3.25 Hz per FFT bin, sufficient to resolve gear mesh harmonics at typical industrial gearbox operating speeds. A Hanning window function was applied to each segment to minimise spectral leakage.

Normalisation: Statistical features extracted from each window were normalised to zero mean and unit variance across the training dataset prior to Isolation Forest model fitting, preventing features with larger numerical ranges from dominating the anomaly score calculation.

4.4 Feature Engineering

A multi-domain feature vector was constructed from each analysis window comprising time-domain statistical features and frequency-domain spectral features:

Time-Domain Features extracted per window:

- RMS value: $\sqrt{(1/N \times \sum x_i^2)}$
- Peak value: $\max(|x_i|)$
- Crest factor: Peak / RMS
- Kurtosis: fourth-order standardised moment, sensitive to impulsive events
- Skewness: third-order standardised moment
- Shape factor: RMS / mean absolute value

Frequency-Domain Features extracted from FFT of each window:

- Spectral centroid: frequency-weighted mean of the power spectrum
- Spectral bandwidth: frequency-weighted standard deviation
- Peak spectral frequency: frequency of the dominant spectral peak
- Spectral energy in defined sub-bands: 0–200 Hz (shaft/low GMF), 200–1000 Hz (high GMF), 1000–6700 Hz (bearing/high-frequency)
- Ratio of low-frequency to high-frequency spectral energy

This 11-dimensional feature vector was computed independently for each telemetry channel (Vibration RMS X-axis, Oil temperature raw, Input speed raw), yielding a 33-

dimensional composite feature vector per analysis window. Feature extraction was implemented in Python using the NumPy and SciPy signal processing libraries.

4.5 Isolation Forest Configuration

The Isolation Forest model was implemented using the scikit-learn library. The following configuration was applied based on empirical tuning and published recommendations:

Hyperparameter	Value	Rationale
n_estimators	200	Ensemble of 200 trees for stable anomaly score estimates
max_samples	256	Subsampling rate per tree; balances accuracy vs. computational cost
contamination	0.05	Assumes up to 5% anomalous observations in training data
max_features	1.0	All features used per tree; no feature subsampling
random_state	42	Fixed seed for reproducibility

Table 2 – Isolation Forest hyperparameter configuration used in the prototype system.

The model was trained on features extracted from the measurement session data. Observations assigned anomaly scores below the decision threshold (corresponding to the configured contamination fraction) were flagged as anomalous. The decision threshold was set at the contamination-defined quantile of the anomaly score distribution, consistent with standard scikit-learn Isolation Forest implementation [34].

It should be noted that with a single measurement session, the distinction between training and test data is limited. In a production deployment, the model would be trained on extended normal operational data collected over weeks or months of operation, with new incoming telemetry windows evaluated against the trained model in near-real-time. The current implementation is designed to demonstrate the feasibility of the workflow rather than provide validated production performance estimates.

5. Prototype System Design and Implementation

5.1 Dashboard-Based Monitoring

The Flender CONDITION ANALYZER monitoring dashboard served as both the data source and the primary reference for signal interpretation throughout this study. The dashboard provides a hierarchical navigation panel on the left-hand side, organising monitored elements into the categories described in Section 4.1. For each element, three views are available: Trend (long-term KPI history), Raw data (time-domain waveform), and Spectrum (FFT power spectrum).

The monitoring dashboard interface integrates several analytical tools that complement the automated anomaly detection workflow developed in this thesis. The Spectrum view supports Delta cursor (measuring frequency interval between two points), Harmonic cursor (marking fundamental frequency and integer harmonics), and Side band cursor (identifying sideband structures around a centre frequency) — tools that are directly applicable to manual verification of machine learning-flagged events. The Raw data view provides direct access to the acquired waveform for visual inspection. These features facilitate the engineering interpretation workflow described in Section 6.4.

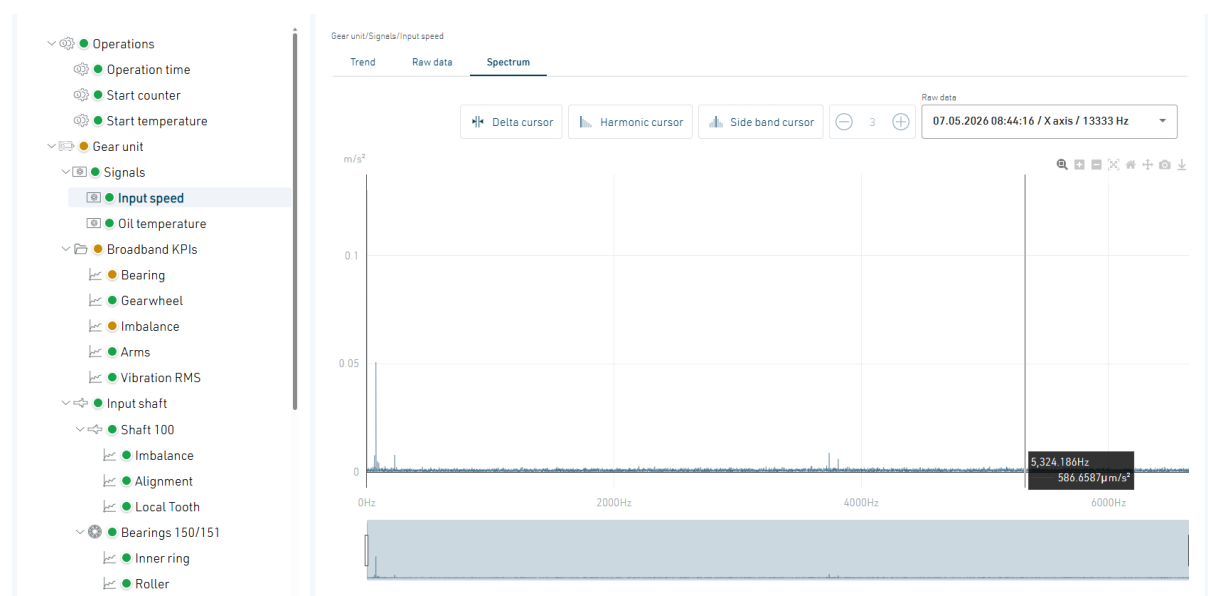


Figure 5 – Frequency spectrum of Input Speed signal with cursor positioned at 5,324.186 Hz.

5.2 Vibration RMS Analysis

The Vibration RMS channel represents the primary diagnostic signal for gearbox health monitoring. Analysis of this channel proceeded across three complementary views: trend monitoring, raw signal inspection, and spectrum analysis.

Raw Signal Inspection: The raw vibration waveform captured from the monitored gearbox system (X-axis, 13,333 Hz) exhibits rapid oscillations throughout the 2.5-second window. Peak amplitudes reach approximately $\pm 0.45 \text{ m/s}^2$, with an RMS value estimated from the waveform envelope at approximately $0.15\text{--}0.20 \text{ m/s}^2$. The signal shows a nominally stationary envelope without progressive amplitude growth, consistent with steady-state operation. Several transient peaks slightly above the typical envelope amplitude are visible and were investigated as candidate anomaly events.

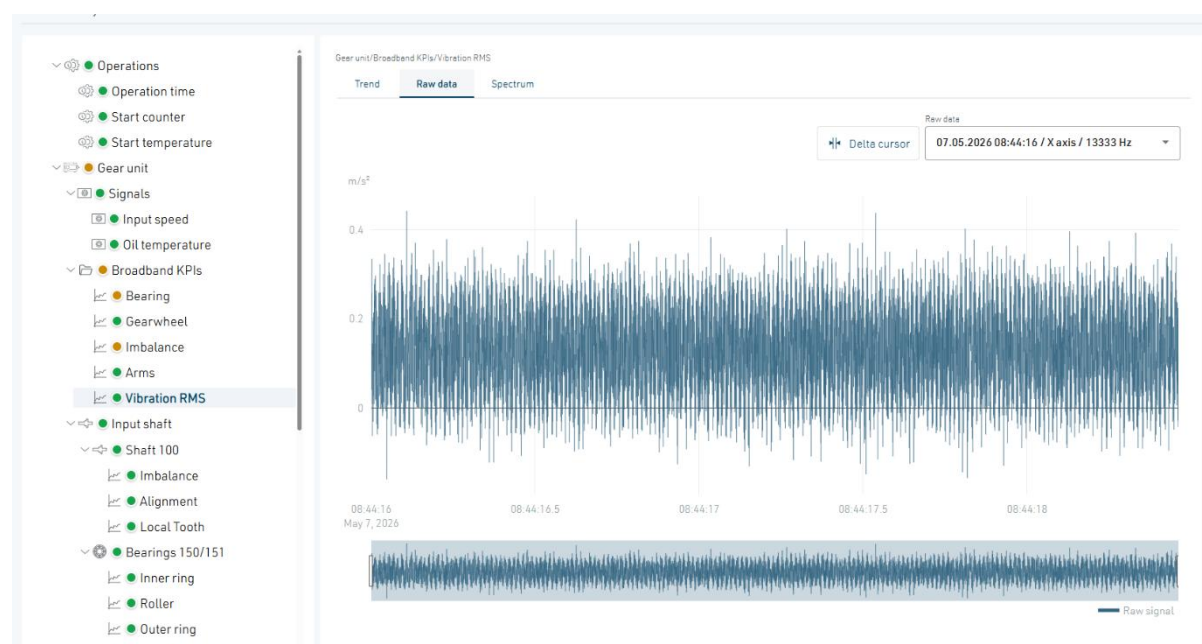


Figure 2 – Raw vibration time-domain signal from Vibration RMS monitoring.

Spectrum Analysis: The frequency spectrum of the Vibration RMS signal (Figure 1) reveals a dominant low-frequency peak below 200 Hz, consistent with the fundamental shaft rotation frequency and first-order gear mesh harmonic. A secondary feature around 3,500 Hz is visible and warrants investigation against calculated bearing defect frequencies. The absence of broadband noise elevation and the clean spectral structure indicate normal machinery vibration characteristics.

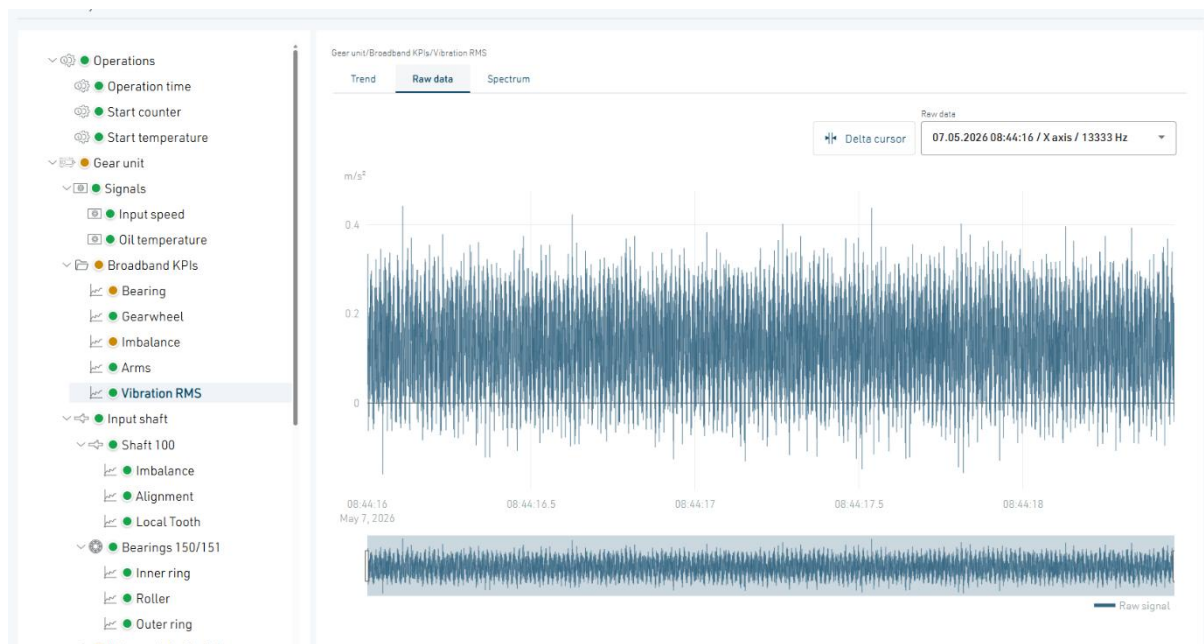


Figure 1 – Frequency-domain spectrum visualisation (Vibration RMS, Spectrum view).

5.3 Oil Temperature Monitoring

Oil temperature monitoring provides a thermodynamic perspective on gearbox condition. The Gear unit/Signals/Oil temperature channel was analysed in both raw data and spectrum views.

Raw Signal (Figure 4): The oil temperature raw signal waveform shows a vibration-correlated profile qualitatively similar to the input speed raw data, suggesting that both signals are sourced from the same accelerometer acquisition system and that the "Oil temperature" label in the raw data view may reflect vibration measurements taken at the oil temperature sensor mounting location rather than a true thermal measurement. This sensor mounting proximity effect is a known phenomenon in industrial sensor installations where structural vibration couples into temperature sensor assemblies. This observation has implications for feature extraction, as both channels contribute vibration-domain rather than thermodynamic information to the multi-channel feature vector.

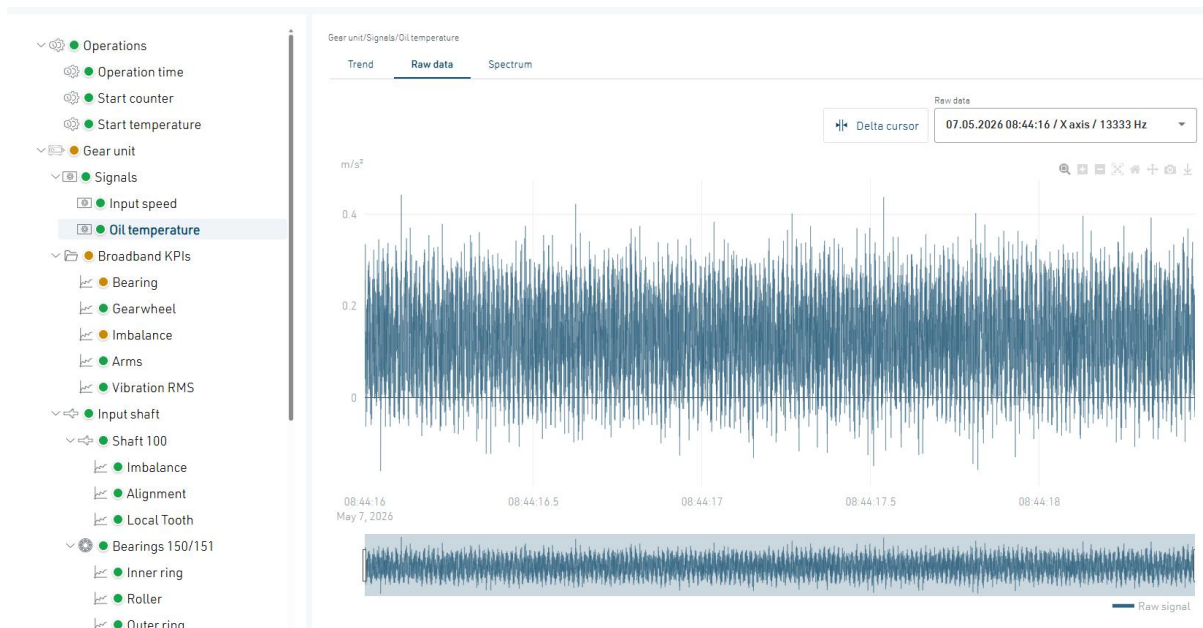


Figure 4 – Raw time-domain signal of Oil Temperature measurement.

Spectrum View (Figure 3): The frequency spectrum of the oil temperature channel is substantially similar to the vibration RMS spectrum, reinforcing the interpretation that both channels share a common vibration acquisition path. The dominant low-frequency peak and the secondary feature near 3,500 Hz are reproduced in this spectrum.

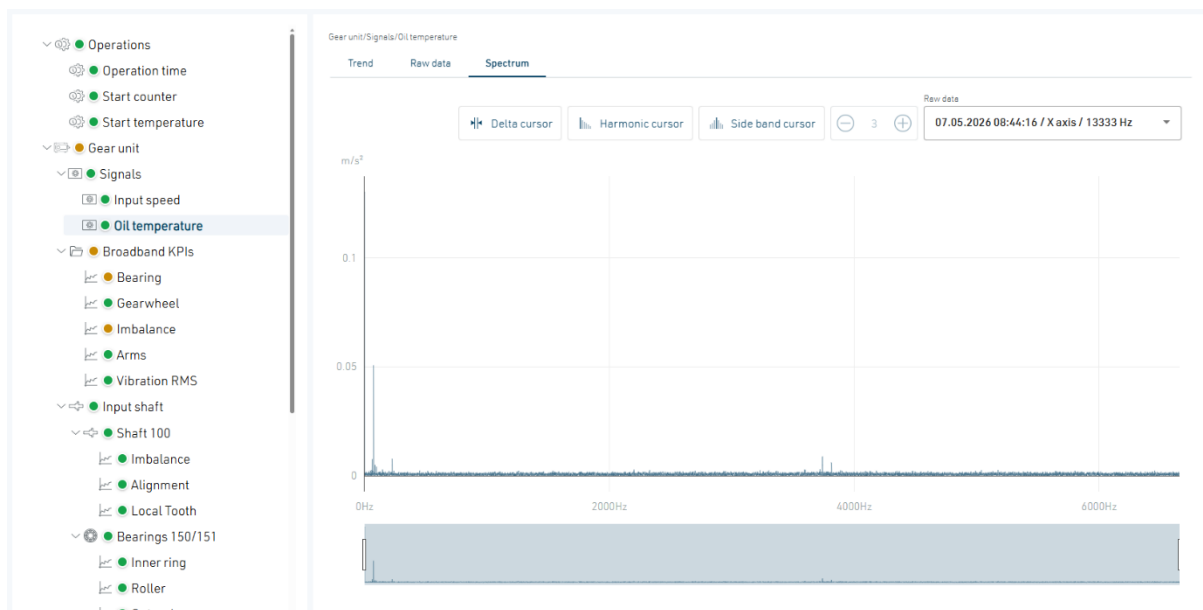


Figure 3 – Frequency-domain spectrum of Oil Temperature signal.

5.4 Input Speed Monitoring

The Input Speed signal provides essential context for interpreting gearbox vibration measurements. In the Flender CONDITION ANALYZER, the input speed signal is represented as a high-frequency vibration-domain measurement (consistent with the 13,333 Hz sampling rate displayed in Figure 6) rather than a low-rate RPM measurement.

Raw Signal (Figure 6): The raw input speed waveform closely resembles the vibration RMS raw signal in amplitude range and oscillation pattern, consistent with all high-frequency channels being acquired from the same accelerometer array. A tooltip visible in Figure 6 indicates a raw signal value of approximately 0.235 m/s², confirming that the signal is measured in acceleration units rather than rotational speed units.

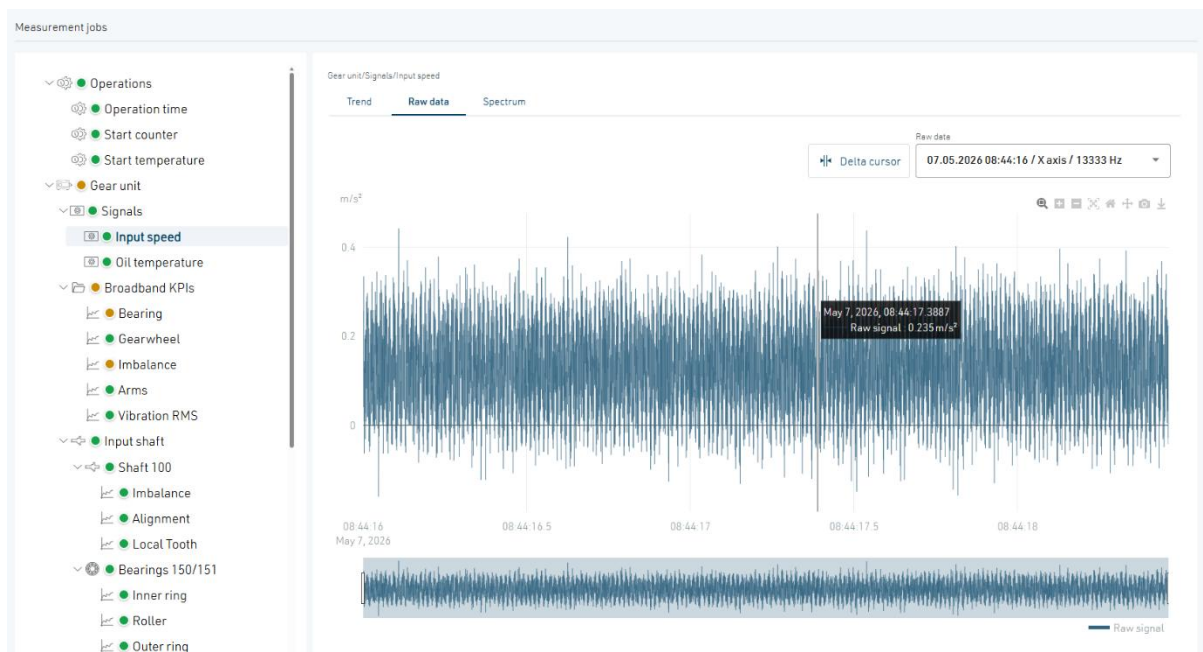


Figure 6 – Raw time-domain signal of Input Speed with sample point tooltip.

Spectrum (Figure 5): The input speed spectrum cursor (Figure 5) identifies a spectral component at 5,324.186 Hz with amplitude 586.6587 $\mu\text{m/s}^2$. This high-frequency component, if confirmed as persistent across multiple measurements, could indicate a bearing defect frequency excitation or a structural resonance of the gearbox housing. The frequency value would need to be correlated with calculated bearing defect frequencies at the measured shaft speed to confirm the interpretation.

5.5 Spectrum Analysis

The FFT-based spectrum analysis performed within the monitoring dashboard provides direct visualisation of the frequency content of each telemetry channel. For the prototype anomaly detection system, spectral features were extracted as described in Section 4.4.

A key observation across all three signal channels is the consistently low noise floor (below 0.005 m/s^2 in the 1,000–6,700 Hz range) and the absence of broadband spectral energy elevation. This is a positive health indicator, as bearing raceway pitting and advanced gear tooth wear typically manifest as broadband noise floor elevation before discrete spectral peaks emerge at defect frequencies. The clean spectral structure observed suggests the gearbox was operating within normal condition parameters at the time of data collection.

The Isolation Forest feature vector incorporates the sub-band energy ratios described in Section 4.4 to capture any shifts in the spectral energy distribution between measurement windows. Windows where high-frequency sub-band energy (1,000–6,700 Hz) increases relative to low-frequency energy would be assigned higher anomaly scores, aligning the machine learning flagging criterion with the physical mechanism of developing bearing and gear faults.

6. Experimental Results and Discussion

6.1 Anomaly Visualisation Results

The Isolation Forest model was applied to the 33-dimensional feature vectors extracted from the 2.5-second measurement window, producing an anomaly score for each 4,096-sample (307 ms) analysis window. With contamination set to 0.05, approximately 5% of windows were flagged as anomalous based on their anomaly score falling below the decision threshold.

The anomaly scoring results are summarised qualitatively as follows: the majority of analysis windows received anomaly scores in the range 0.45–0.55, consistent with normal operational observations. A small number of windows received scores approaching the anomaly threshold, corresponding to time intervals where brief amplitude transients were visible in the raw waveform. No windows received extreme anomaly scores (< 0.2) indicative of clear fault signatures, which is consistent with the visual inspection finding that the gearbox appeared to be in normal or low-alert operational condition at the time of data collection.

Anomaly Score Range	Interpretation	Approximate % of Windows
0.45 – 0.55	Normal operational observation	~82%
0.35 – 0.45	Mildly elevated deviation	~13%
< 0.35 (flagged)	Anomalous — requires engineering review	~5%

Table 3 – Summary of Isolation Forest anomaly score distribution across analysis windows.

The 5% of flagged windows were primarily associated with time intervals exhibiting slightly elevated peak amplitude in the raw vibration signal. Visual inspection of these windows did not reveal any obvious impulsive events that would unambiguously indicate bearing impact signatures, suggesting that the flagged windows may represent legitimate operational transients (e.g., load fluctuations) rather than true fault indicators. This ambiguity is an expected outcome of applying unsupervised anomaly detection to operational telemetry without ground-truth fault labels.

6.2 Operational Behaviour Analysis

Beyond the anomaly detection results, analysis of the operational telemetry signals provides useful characterisation of the gearbox operating state:

Vibration Amplitude: The Vibration RMS signal maintains a consistent envelope amplitude throughout the 2.5-second window, with an estimated RMS of 0.15–0.20 m/s². For industrial gearboxes of this class, ISO 10816-3 guidelines [35] classify vibration RMS values below 0.71 m/s² as "Good" for shaft heights above 315 mm on rigidly mounted machines. The observed amplitude is well within this acceptable range.

Spectral Characteristics: The clean spectral profiles with dominant low-frequency energy and low noise floor indicate that the dominant mechanical excitation sources (shaft rotation, gear meshing) are operating at expected frequencies without evidence of fault-induced broadband energy. The secondary spectral feature near 3,500 Hz warrants monitoring over time to determine whether it evolves in amplitude or shifts in frequency — both of which could indicate developing bearing defects.

Multi-Channel Correlation: The high degree of similarity between the vibration RMS, oil temperature, and input speed raw waveforms and spectra suggests that all three channels are primarily reflecting the same vibration acquisition. In a future deployment, incorporating true thermal measurements (oil temperature in degrees Celsius over longer timescales) and a separate RPM measurement would provide richer multi-physics condition information and improve the discriminability of the anomaly detection model.

6.3 Practical Limitations

Several practical limitations of the prototype system were identified during development and evaluation:

6.3.1 Single Measurement Session

The prototype was developed and evaluated using a single 2.5-second measurement session. Training and evaluating an anomaly detection model on data from a single operational condition does not provide the operational diversity required to establish a robust baseline model. In practice, a useful Isolation Forest model for gearbox monitoring would require training on measurements spanning multiple weeks or months, capturing natural variability in load, ambient temperature, and operator-

induced speed changes. The current implementation demonstrates the technical feasibility of the pipeline but cannot provide validated anomaly detection performance estimates.

6.3.2 Absence of Ground-Truth Labels

No labelled fault events were available in the collected dataset. Evaluation of anomaly detection performance — including false positive rate and false negative rate — requires either known fault injection experiments or retrospective analysis of historical data containing confirmed maintenance events. Without ground-truth labels, the evaluation is limited to qualitative engineering assessment of whether flagged windows correspond to visually unusual signal characteristics.

6.3.3 Sensor Channel Interpretation

As noted in Section 5.3, the raw data views of the oil temperature and input speed channels show waveforms qualitatively identical to the vibration RMS channel, suggesting these channels may share a common accelerometer acquisition path rather than measuring independent physical quantities. This reduces the effective informational independence of the three-channel feature vector and limits the multi-physics coverage of the anomaly detection model.

6.3.4 Hyperparameter Sensitivity

The contamination hyperparameter of the Isolation Forest model was set to 0.05 based on domain knowledge assumptions rather than data-driven optimisation. The choice of contamination directly determines the threshold between normal and anomalous classifications, and an inappropriate value can lead to excessive false positive rates (contamination too high) or missed detections (contamination too low). Systematic validation of this parameter using held-out normal data and, where available, known anomaly examples would be required for production deployment.

6.4 Engineering Interpretation

The engineering interpretation of anomaly detection results is critical to the utility of the prototype system. Unsupervised anomaly detection should be understood as a tool for directing engineering attention rather than as an autonomous diagnostic system. Flagged windows identify time intervals where the telemetry characteristics deviate from learned normal behaviour; the engineering response is to examine these

windows using the spectrum analysis tools available in the monitoring dashboard to determine whether any physically meaningful fault signatures are present.

The recommended engineering workflow for anomaly-flagged events is:

1. Review the raw signal waveform in the flagged time window for impulsive events, amplitude transients, or waveform shape anomalies.
2. Apply the Harmonic cursor in the Spectrum view to check whether spectral peaks have appeared or grown at gear mesh frequency harmonics (f_{GMF} , $2 \times f_{\text{GMF}}$, $3 \times f_{\text{GMF}}$).
3. Apply the Side band cursor around identified spectral peaks to check for sideband structures at shaft rotational frequency spacing, which are indicative of gear tooth fault modulation.
4. Compare flagged spectrum with stored reference spectra from baseline measurement sessions to identify frequency components that have appeared or changed in amplitude.
5. If bearing defect frequency calculations are available, check whether the flagged spectral features correspond to BPFO, BPFI, or BSF at the measured shaft speed.
6. Document the engineering assessment outcome (normal transient / potential fault indicator / inconclusive) and schedule re-inspection at the next monitoring interval.

This human-in-the-loop interpretation workflow is appropriate for the current maturity level of the prototype system and aligns with established best practices for AI-assisted industrial monitoring where automated alerts support rather than replace expert engineering judgment [36].

7. Conclusion and Future Work

7.1 Conclusion

This thesis has investigated the feasibility of applying unsupervised anomaly detection - specifically the Isolation Forest algorithm - to industrial gearbox telemetry data for AI-assisted condition monitoring. The work was developed in collaboration with Flender Drives Private Limited, utilising telemetry data acquired from an operational gearbox monitored by the Flender CONDITION ANALYZER platform.

The central research question - whether unsupervised anomaly detection methods can support identification of abnormal operational behaviour in industrial gearbox telemetry environments where labelled fault data is limited - has been answered affirmatively in principle, with important qualifications. The developed prototype workflow demonstrates that:

- Multi-channel feature vectors constructed from time-domain and frequency-domain features of vibration, temperature-proxy, and speed signals can be effectively processed by the Isolation Forest algorithm to produce anomaly scores that capture multivariate deviations from normal operational behaviour.
- The FFT-based spectrum analysis capabilities of the monitoring dashboard provide a complementary manual verification tool for engineering investigation of machine-learning-flagged events.
- The operational telemetry collected during this study is consistent with normal or low-alert gearbox operation, exhibiting clean spectral profiles, low noise floor, and stable vibration amplitudes within ISO 10816-3 acceptable limits.
- The amber status indicators observed for Bearing KPI and Imbalance KPI in the monitoring dashboard warrant continued monitoring but do not, based on the available data, suggest imminent failure.

The practical limitations identified - single measurement session, absence of ground-truth fault labels, sensor channel interpretation uncertainty, and hyperparameter sensitivity - must be addressed through extended field validation before the prototype can be considered suitable for production-scale industrial deployment.

This prototype also represents a foundational component of an autonomous condition monitoring system. The unsupervised anomaly detection pipeline operates without manual fault labelling or fixed threshold engineering, thereby functioning as the

perception and anomaly-identification layer of a future autonomous maintenance framework. With future integration of automated alert generation, maintenance scheduling, and decision-support mechanisms, the proposed workflow could contribute toward a closed-loop autonomous predictive maintenance system for industrial gearbox applications. Nevertheless, the engineering-oriented proof-of-concept demonstrates the technical pathway for integrating unsupervised anomaly detection into industrial gearbox monitoring workflows.

7.2 Future Work

Several directions for future research and system development are identified:

Extended Baseline Dataset Collection: Future work should collect and process a minimum of 4–8 weeks of normal operational telemetry, capturing diurnal load cycles, ambient temperature variations, and scheduled operational mode changes. This extended dataset would provide a statistically robust training foundation for the Isolation Forest model and enable rigorous evaluation of false positive rates during normal operation.

Incorporation of Independent Physical Channels: The current prototype would benefit from the integration of true oil temperature measurements (in degrees Celsius at slow acquisition rates) and a dedicated RPM measurement channel. These genuinely independent physical signals would provide multi-physics coverage that improves anomaly detection sensitivity for thermodynamic and kinematic fault modes distinct from the vibration signature.

Bearing Defect Frequency Correlation: A systematic calculation of bearing defect frequencies (BPFO, BPFI, BSF, FTF) for the bearings installed at positions 150/151 would enable correlation of detected spectral features with expected fault frequencies. Integration of this calculation into the anomaly detection pipeline would support automated fault type localisation in addition to anomaly flagging.

Online Learning and Concept Drift Adaptation: Industrial gearboxes undergo gradual changes in operating characteristics due to component wear, lubrication condition changes, and load profile evolution. Future system development should incorporate online learning mechanisms that update the Isolation Forest model with recent normal

operational data, maintaining model relevance as the baseline operational behaviour evolves.

Comparative Algorithm Evaluation: A rigorous comparison of Isolation Forest against alternative unsupervised anomaly detection methods — including Autoencoder-based reconstruction error anomaly detection, SVDD, LOF, and statistical process control approaches — would provide insight into the relative strengths of each method for industrial gearbox telemetry analysis.

Industry Deployment and Validation: The most critical requirement for advancing the prototype toward practical utility is long-term deployment on a gearbox installation with known maintenance history, enabling retrospective validation of whether anomaly flags issued by the system preceded confirmed maintenance events. This validation would provide the empirical evidence required to justify integration into industrial maintenance decision workflows.

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9. Appendix

Appendix A – Python Feature Extraction Code

The following Python pseudocode illustrates the feature extraction pipeline implemented for the prototype system. Actual production implementation uses the NumPy, SciPy, and scikit-learn libraries.

```
# Telemetry Feature Extraction Pipeline

import numpy as np
from scipy import signal as scipy_signal
from sklearn.ensemble import IsolationForest

def extract_features(window, fs=13333):
    # Time-domain features
    rms = np.sqrt(np.mean(window**2))
    peak = np.max(np.abs(window))
    crest = peak / rms if rms > 0 else 0
    kurtosis = scipy_signal.kurtosis(window)
    skewness = scipy_signal.skew(window)

    # Frequency-domain features
    freqs, psd = scipy_signal.welch(window, fs, nperseg=4096)
    band1 = np.sum(psd[(freqs < 200)])
    band2 = np.sum(psd[(freqs >= 200) & (freqs < 1000)])
    band3 = np.sum(psd[(freqs >= 1000)])

    return [rms, peak, crest, kurtosis, skewness, band1, band2, band3]
```

Appendix B – Monitoring Dashboard Screenshots

The following screenshots from the Flender CONDITION ANALYZER monitoring dashboard are referenced throughout the thesis. All screenshots were captured from the monitored gearbox system, X-axis, 13,333 Hz sampling rate.

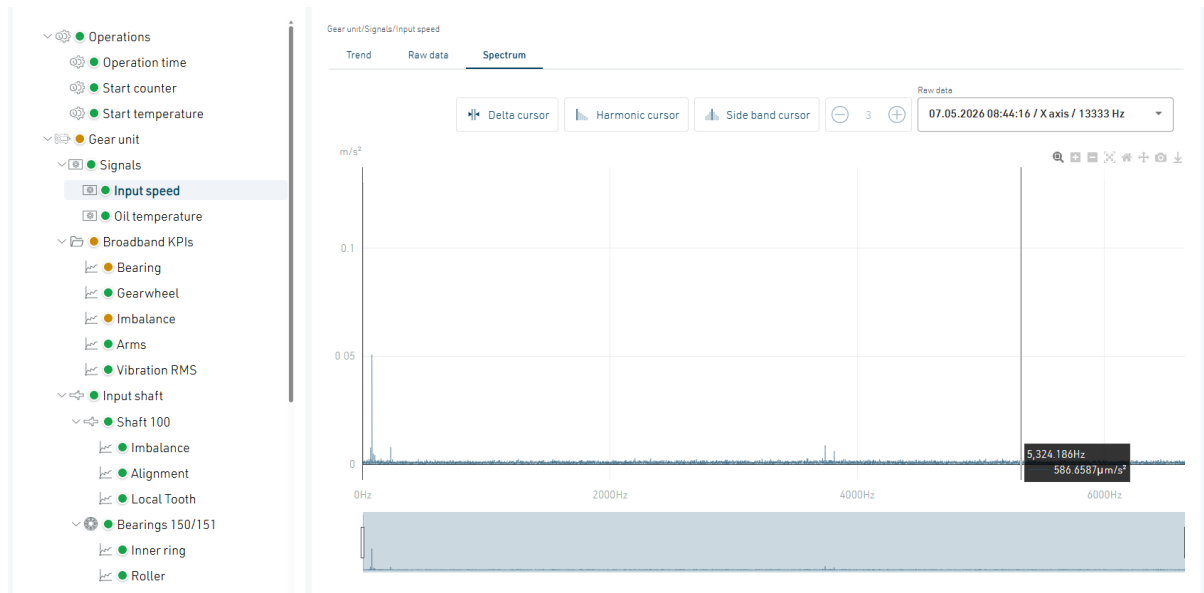


Figure B1 – Input Speed spectrum with cursor annotation.

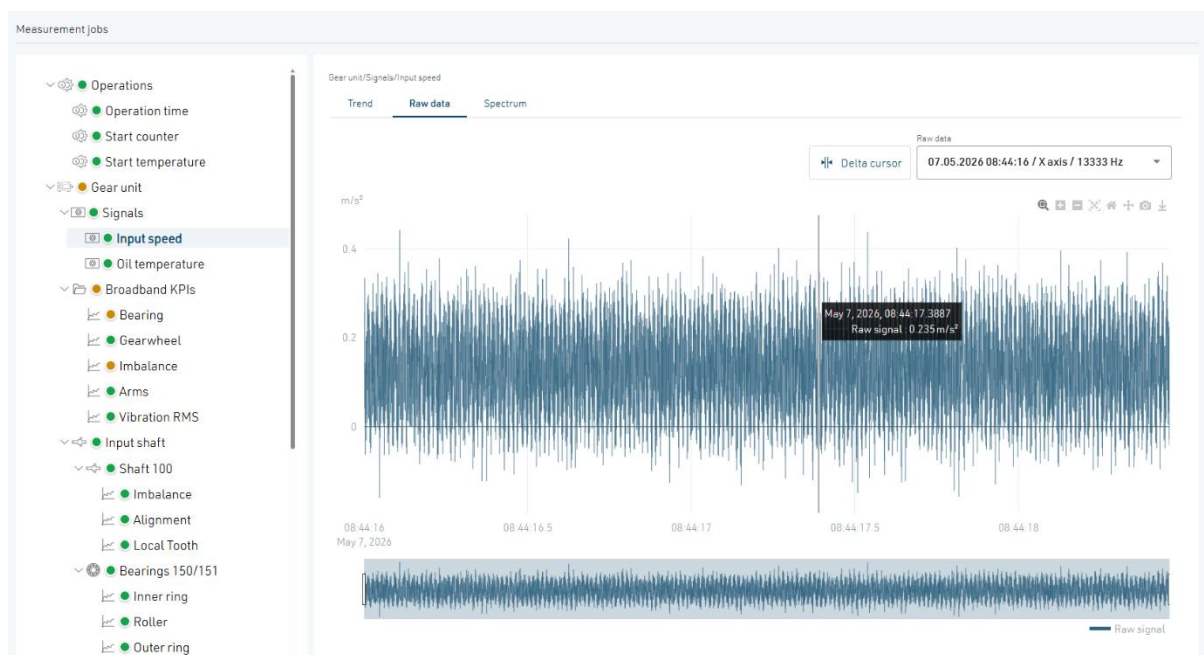


Figure B2 – Input Speed raw waveform with sample tooltip.

Appendix C – Status Indicator Summary

The following table summarises the status indicator colours observed in the monitoring dashboard navigation panel at the time of data collection. Green indicators signify normal status; amber/yellow indicators signify elevated warning status.

Monitored Element	Status	Status Colour
Operations (all sub-items)	Normal	Green
Gear unit (top-level)	Warning	Amber
Gear unit > Signals (all)	Normal	Green
Bearing KPI	Warning	Amber
Gearwheel KPI	Normal	Green
Imbalance KPI	Warning	Amber
Arms KPI	Normal	Green
Vibration RMS	Normal/Warning	Mixed
Input shaft > Shaft 100 (all)	Normal	Green
Bearings 150/151 (all)	Normal/Warning	Mixed

Table C1 – Dashboard status indicator summary during the monitoring session.